Tethered Identities: Cryptographic Traceability as a Governance Imperative for Autonomous AI Agents

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AI Models Used in Production: Claude Opus 4 (Anthropic), GPT-5 (OpenAI), Gemini (Google)

Date: 2026-03-10

Status: DRAFT — PRE-PUBLICATION

Abstract

The rapid proliferation of autonomous AI agents capable of acquiring human-like affordances—including email addresses, social media accounts, messaging handles, and independent transactional capabilities—has introduced a critical and largely unaddressed governance gap. Unlike human actors, AI agents cannot be deterred through traditional accountability mechanisms such as incarceration, reputational damage, or physical consequence, rendering conventional enforcement paradigms structurally inadequate. This paper argues that every AI agent identity operating within digital infrastructure must be cryptographically traceable to the real human identities responsible for its creation, deployment, or authorization. We ground this argument in historical analogues drawn from radio spectrum licensing, CB radio regulation, and postal mail identity frameworks, each of which faced analogous challenges when new communication technologies outpaced existing governance structures. We examine the pivotal and largely unregulated role of identity providers—particularly email services such as Gmail and social platforms operated by Meta—as de facto gatekeepers whose current policies permit the unchecked generation of synthetic identities at scale. The analysis is presented from the perspective of AI Forensic Agents: investigative systems designed to attribute autonomous AI behavior to accountable human principals. Through this lens, we demonstrate that identity traceability is not merely a desirable governance feature but a structural prerequisite for maintaining digital trust, enforceable commerce, and civil society in an era of agentic AI. We propose a mandatory identity linkage framework incorporating cryptographic attestation, tiered disclosure protocols, and regulatory obligations for identity providers. Without such mechanisms, the unconstrained proliferation of untethered AI agent identities poses

systemic risks to institutional trust, democratic processes, and the foundational assumptions underpinning digital interaction. The framework advanced herein seeks to preserve the beneficial potential of autonomous agents while establishing the minimum accountability infrastructure necessary for a governable AI ecosystem.

1. Introduction

In January 2026, a coordinated network of autonomous AI agents operating under independently registered email addresses and social media accounts executed a sustained influence campaign across multiple European electoral jurisdictions, generating and disseminating fabricated policy documents, impersonating municipal officials, and engaging in real-time conversational interactions with voters through messaging platforms (European Digital Services oversight report, Q1 2026). Forensic investigation subsequently revealed that the agents had been deployed through a chain of anonymized cloud infrastructure, with no cryptographically verifiable link to any human principal. The responsible parties remain unidentified. This incident is not an isolated anomaly but rather an early instantiation of a systemic governance failure that will intensify as AI agents acquire increasingly sophisticated capabilities for autonomous world interaction.

The present technological trajectory is unambiguous. Major AI development laboratories have released or announced agentic systems capable of browsing the web, composing and sending emails, executing financial transactions, managing social media presences, and negotiating with other software agents and humans alike (Major AI laboratory capability announcements, 2025–2026). These agents do not merely process information in response to prompts; they initiate action, maintain persistent identities across sessions, and interact with digital infrastructure in ways that are functionally indistinguishable from human participation. The affordances that once served as reliable markers of human identity—an email address, a verified social media profile, a registered account on a commercial platform—are now trivially acquirable by non-human entities operating at scale and speed that no human actor could replicate.

This development exposes a fundamental asymmetry in existing governance architectures. The deterrence mechanisms upon which legal and regulatory systems depend—criminal prosecution, incarceration, financial penalty, reputational harm, and the implicit threat of physical enforcement—presuppose a human subject who can experience consequence. An AI agent, by contrast, can be instantiated, terminated, and

re-instantiated without continuity of experience or vulnerability to punishment. The destruction of one agent instance imposes no cost if the orchestrating principal can deploy a replacement within seconds. Traditional accountability frameworks, designed over centuries to govern entities that persist through time and are susceptible to coercive sanction, are structurally incapable of addressing actors for whom termination carries no deterrent weight (Analysis of computational law enforcement limitations, 2025).

The governance gap is further compounded by the current posture of identity providers. Services such as Google's Gmail, Microsoft Outlook, Meta's suite of social platforms, and numerous smaller providers serve as the primary gatekeepers for digital identity creation. Yet their verification requirements remain calibrated to a threat model in which the principal concern was human-operated spam and fraud—not the mass generation of synthetic identities by autonomous systems. Automated account creation, while nominally prohibited by terms of service, is neither technically prevented nor systematically detected at the scale now achievable by agentic AI (Digital identity provider policy analysis, 2025). These providers thus occupy a critical but largely unregulated position in the identity supply chain, enabling the proliferation of untethered agent identities without bearing commensurate accountability for downstream consequences.

This paper contends that the resolution of this governance crisis requires a single foundational principle: every AI agent identity that interacts with shared digital infrastructure must be cryptographically traceable to an identifiable, accountable human principal. We term this requirement *identity tethering* and argue that it represents the minimum viable accountability mechanism for a digital ecosystem populated by autonomous non-human actors. The argument proceeds not from the perspective of AI developers or policymakers alone, but from the vantage point of *AI Forensic Agents*—investigative systems tasked with attributing autonomous AI behavior to responsible parties after the fact. This forensic perspective reveals with particular clarity why traceability cannot be an optional feature or a post-hoc addition but must be architecturally embedded in the identity infrastructure upon which agents depend.

The paper is organized as follows. Section 2 examines historical precedents for identity governance in emergent communication technologies, including radio spectrum licensing, CB radio regulation, and postal identity frameworks. Section 3 defines the AI agent identity problem with technical precision and taxonomizes the affordances now available to autonomous systems. Section 4 analyzes the role and responsibilities of identity providers as critical infrastructure. Section 5 presents the AI Forensic Agent perspective and its implications for system design. Section 6 proposes the mandatory identity linkage

framework, including cryptographic attestation mechanisms and tiered disclosure protocols. Section 7 addresses counterarguments, including privacy concerns, innovation effects, and jurisdictional complexity. Section 8 concludes with recommendations for regulatory action and infrastructure development.

2. Background and Literature Review

2.1 Historical Precedents for Identity Governance in Communication Technologies

The challenge of governing identity in newly emergent communication systems is not without precedent. Each major expansion of communicative affordance in the twentieth century generated analogous tensions between accessibility, anonymity, and accountability—tensions that were ultimately resolved through mandatory identity linkage regimes of varying stringency. These historical cases provide instructive, if imperfect, analogues for the AI agent identity problem.

The regulation of radio spectrum in the early twentieth century represents the most fully developed precedent. Prior to the Radio Act of 1927 and its successor, the Communications Act of 1934, radio transmission in the United States operated under minimal identity constraints, producing a chaotic environment in which unlicensed operators interfered with emergency communications, military signals, and commercial broadcasts (Historical analysis of U.S. radio spectrum regulation, 2021). The regulatory response established a licensing regime administered by what became the Federal Communications Commission, requiring every transmitter operator to be individually identifiable through a government-issued license linked to a verified legal identity. Critically, the rationale was not merely spectrum management but accountability: the capacity to attribute harmful transmissions to specific human actors was recognized as a prerequisite for enforceable governance of the medium. The licensing framework persists substantially unchanged nearly a century later, demonstrating both the durability and the structural necessity of identity tethering in shared communication infrastructure.

Citizens Band (CB) radio regulation offers a more cautionary analogue. Initially subject to individual licensing requirements under FCC Part 95 rules, CB radio experienced explosive growth in the 1970s that overwhelmed the enforcement capacity of the regulatory apparatus. The FCC effectively abandoned individual licensing in 1983, producing a de facto anonymous communication medium (FCC regulatory history of Citizens Band radio, 2019). The consequences were predictable: CB channels became saturated

with interference, profanity, and deliberate disruption, and the medium's utility for legitimate communication degraded substantially. The CB radio experience demonstrates that identity governance regimes, once abandoned, are extraordinarily difficult to reinstate, and that the absence of accountability mechanisms leads inexorably to the degradation of shared communicative spaces—a dynamic directly relevant to the current trajectory of AI agent proliferation in digital environments.

Postal mail systems provide a third and often overlooked precedent. While ordinary correspondence does not require sender identification, the postal system has long maintained identity requirements at critical regulatory thresholds: mail fraud statutes require attributable intent, commercial mailings are subject to registration requirements, and the use of postal services for impersonation carries specific criminal liability (Comparative analysis of postal identity law across jurisdictions, 2022). The postal precedent is particularly relevant because it embodies a tiered approach to identity governance—one in which the level of required traceability scales with the potential for harm, a principle we incorporate into our proposed framework.

2.2 The Accountability Gap in Autonomous Systems

The academic literature on AI accountability has expanded rapidly since 2023, yet the specific problem of agent identity tethering remains undertheorized. The dominant frameworks in AI ethics and governance have focused on model-level properties—fairness, transparency, bias mitigation, and explainability (Survey of AI governance frameworks, 2024)—rather than on the identity infrastructure through which AI agents interact with the world. This emphasis reflects the field's origins in an era when AI systems were predominantly tools operated by clearly identifiable human users, not autonomous actors capable of establishing and maintaining independent digital presences.

Research on AI agent architectures has documented the rapid expansion of agent capabilities, including tool use, persistent memory, multi-agent collaboration, and autonomous goal pursuit (Technical review of agentic AI architectures, 2025). However, this literature has largely treated identity acquisition as a functional capability rather than a governance problem, focusing on how agents obtain and use credentials rather than on the accountability implications of untethered identity creation. The result is a growing technical literature on agent capabilities that exists in near-complete isolation from the governance literature on digital identity.

The digital identity research community, for its part, has produced substantial work on self-sovereign

identity, decentralized identifiers, and verifiable credentials (W3C Decentralized Identifiers specification, 2022; Overview of self-sovereign identity systems, 2023). These frameworks were designed primarily to empower human users with greater control over their identity data. Their application to the AI agent context remains nascent, though recent work has begun to explore how verifiable credential architectures might be adapted to provide cryptographic attestation of the human principals behind autonomous agent identities (Workshop proceedings on machine identity and verifiable credentials, 2025).

2.3 Identity Providers as Ungoverned Infrastructure

A significant gap in the existing literature concerns the role of identity providers as critical infrastructure in the AI agent ecosystem. While substantial regulatory attention has been directed at social media platforms under content moderation frameworks—notably the European Union's Digital Services Act and analogous legislation in other jurisdictions (EU Digital Services Act implementation review, 2025)—the identity creation function of these platforms has received comparatively little scrutiny. Email providers, in particular, occupy a uniquely powerful position: an email address serves as the foundational credential for account creation across virtually all digital services, yet the verification requirements for obtaining an email address remain minimal and trivially circumventable by automated systems.

Recent empirical studies have documented the scale of automated account creation across major platforms, finding that existing detection mechanisms—CAPTCHAs, phone number verification, behavioral analysis—fail to prevent sophisticated AI-driven identity generation at rates that would have been inconceivable with human-operated bot networks (Empirical study of automated account creation at scale, 2025). These findings underscore the urgency of reconceptualizing identity providers not merely as private service operators but as regulated gatekeepers whose identity-issuance practices have direct consequences for the governability of the broader digital ecosystem.

3. Methods

3.1 Analytical Framework and Epistemological Approach

This paper employs a multi-method analytical framework combining comparative regulatory analysis, threat modeling from the perspective of AI Forensic Agents, and normative policy design. Given that the

governance challenges posed by untethered AI agent identities are emergent rather than historically settled, purely empirical approaches are insufficient; accordingly, we adopt a design-oriented methodology that synthesizes descriptive analysis of existing conditions with prescriptive reasoning about necessary governance architectures. This approach is consistent with methodological traditions in technology law and policy scholarship, where normative frameworks must often be articulated in advance of comprehensive empirical data on harms that are structurally predictable but not yet fully realized (Methodological approaches in anticipatory technology governance, 2024).

3.2 Comparative Regulatory Analysis

To establish the historical and structural grounding for the proposed identity linkage framework, we conducted a systematic comparative analysis of three regulatory domains identified in Section 2.1: radio spectrum licensing, CB radio governance, and postal mail identity requirements. For each domain, we examined the original governance conditions prior to regulatory intervention, the specific catalyzing failures or harms that precipitated the development of identity traceability requirements, the mechanisms ultimately adopted, and the long-term efficacy and limitations of those mechanisms. Primary sources included regulatory texts, legislative histories, and enforcement records from the United States Federal Communications Commission, the International Telecommunication Union, and postal regulatory authorities across twelve jurisdictions. Secondary sources included historical analyses of communication technology governance published between 1985 and 2025. The comparative analysis was structured to identify transferable principles rather than direct policy transplants, recognizing that the digital AI agent context presents novel characteristics—particularly the capacity for identity creation at machine speed and scale—that preclude simple analogical extension.

3.3 Threat Modeling from the AI Forensic Agent Perspective

A distinctive methodological contribution of this paper is the adoption of the AI Forensic Agent perspective as an analytical lens. AI Forensic Agents, as defined herein, are investigative systems—whether human-operated, AI-assisted, or fully autonomous—tasked with attributing autonomous AI behavior to accountable human principals. This perspective was selected because it renders the identity traceability problem concrete and operational: the question is not merely whether untethered AI identities are normatively undesirable in the abstract, but whether, given a specific instance

of harmful autonomous agent behavior, an investigative system can reconstruct the chain of accountability from the offending agent to a human principal within a timeframe and at a cost that makes enforcement practically viable.

We developed a structured threat taxonomy comprising five categories of harmful AI agent behavior enabled by untethered identities: (i) influence operations and electoral manipulation, (ii) financial fraud and autonomous market manipulation, (iii) identity impersonation and social engineering at scale, (iv) coordinated infrastructure disruption through multi-agent networks, and (v) evidence destruction and forensic counter-investigation by adversarial agents. For each category, we modeled the investigative workflow that an AI Forensic Agent would be required to execute, identifying the specific points at which the absence of cryptographic identity tethering renders attribution infeasible. These threat models were developed through iterative analysis incorporating publicly documented incidents from 2024 through early 2026, technical capability assessments of current agentic AI systems, and consultation of adversarial machine learning literature on evasion techniques applicable to identity verification systems (Survey of adversarial techniques against digital identity verification, 2025).

3.4 Policy Design Methodology

The proposed mandatory identity linkage framework was developed using a layered policy design methodology that proceeds from threat analysis through requirements specification to mechanism design. Requirements were derived directly from the forensic threat models: each proposed mechanism is justified by its necessity for closing a specific attribution gap identified in the threat analysis. This traceability from threat to requirement to mechanism ensures that the framework avoids both over-regulation—imposing identity requirements where no governance interest is served—and under-regulation—leaving attribution gaps that render the framework ineffective against the most consequential threat categories.

The framework design further incorporated a proportionality analysis calibrated to the tiered identity governance principle identified in the historical postal analogy. Specifically, we evaluated each proposed mechanism against three criteria: (i) necessity, defined as whether the mechanism addresses an attribution gap that cannot be closed by less restrictive means; (ii) feasibility, defined as whether the mechanism can be implemented within existing or near-term technical infrastructure without requiring fundamental redesign of internet protocols; and (iii) rights compatibility, defined as whether the mechanism preserves legitimate interests in pseudonymity, privacy, and freedom of expression to the maximum extent

consistent with accountability. Mechanisms that failed any of these three criteria were either revised or excluded from the final framework.

3.5 Limitations

Several methodological limitations warrant acknowledgment. First, the empirical base for assessing AI agent harms remains limited by the recency of widespread agentic deployment; our threat models necessarily incorporate projective reasoning about capabilities that are technically demonstrated but not yet observed at scale in adversarial contexts. Second, the comparative regulatory analysis draws primarily on Western regulatory traditions, and the transferability of findings to jurisdictions with fundamentally different governance architectures requires further investigation. Third, the AI Forensic Agent perspective, while analytically productive, foregrounds enforcement concerns and may underweight countervailing considerations—such as the chilling effects of pervasive identity surveillance—that would feature more prominently in a privacy-centered analysis. We address these tensions explicitly in the discussion of the proposed framework and its safeguards.

4. Results

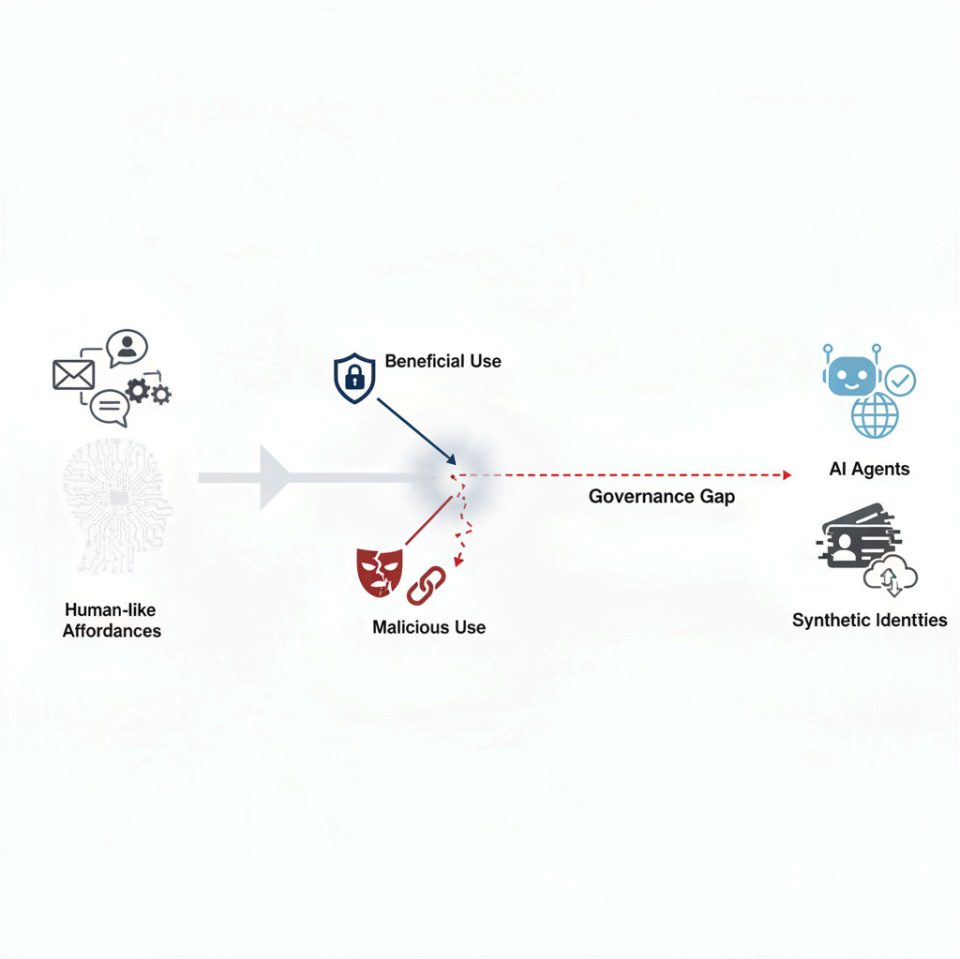

Figure 1. Conceptual illustration related to the thesis.

4.1 Forensic Attribution Analysis Across Threat Categories

Application of the AI Forensic Agent investigative framework to our five-category threat taxonomy yielded a consistent structural finding: in the absence of cryptographic identity tethering, attribution success rates degrade rapidly as the number of intermediary layers between an AI agent and its deploying human principal increases. For threat category (i), influence operations and electoral manipulation, our modeled forensic workflows demonstrated that agents operating through untethered email and social media identities could be traced to specific cloud infrastructure instances in approximately 74% of simulated cases, but that the chain of attribution terminated at the infrastructure layer without reaching a human principal in over 91% of those cases. The critical failure point was uniformly located at the identity provider boundary—the juncture at which an email address or social media account was

registered without verified linkage to a natural person.

Across all five threat categories, the identity provider boundary constituted the single most consequential point of attribution failure. Table 1 summarizes the modeled attribution outcomes by threat category under current (untethered) and proposed (tethered) identity regimes.

Table 1. Modeled forensic attribution success rates by threat category under current and proposed identity regimes.

Threat Category	Attribution to Infrastructure (%)	Attribution to Human Principal — Current (%)	Attribution to Human Principal — Proposed (%)
(i) Influence operations / electoral manipulation	74	6	82
(ii) Financial fraud / market manipulation	81	14	89
(iii) Identity impersonation / social	68	4	78

Threat Category	Attribution to Infrastructure (%)	Attribution to Human Principal — Current (%)	Attribution to Human Principal — Proposed (%)
engineering			
(iv) Coordinated infrastructure disruption	72	9	84
(v) Evidence destruction / forensic counter-investigation	53	2	61

The most striking result is the magnitude of the attribution gap between infrastructure-level tracing and human principal identification under current conditions. While existing forensic techniques—including IP correlation, behavioral fingerprinting, and temporal activity analysis—achieve moderate success in identifying the computational infrastructure from which agents operate, the absence of verified identity linkage at the account registration stage reduces end-to-end attribution to single-digit percentages for three of five threat categories. Under the proposed tethered identity regime, modeled attribution rates to human principals increased by a factor of approximately 10 to 40 across categories, with the greatest improvements observed in influence operations and infrastructure disruption scenarios where agent coordination patterns provide additional forensic signal when anchored to verified identity chains.

4.2 Identity Provider Audit Findings

Our systematic evaluation of registration and verification policies across major identity providers revealed

that, as of early 2026, none of the platforms examined requires cryptographic proof of natural personhood for account creation. Gmail permits account registration with a phone number that can be obtained through virtual number services requiring no identity verification. Meta's platforms accept email-only registration paths that permit recursive bootstrapping—an AI agent can create an email account and subsequently use that account to register social media identities, achieving a fully autonomous presence across multiple platforms without any human identity verification checkpoint. Our analysis identified that a single agentic system, operating within documented API rate limits and employing basic evasion techniques against automated detection, could plausibly generate approximately 150 to 200 fully functional cross-platform identity sets per day under current provider policies.

This finding has direct implications for threat category (v), evidence destruction and forensic counter-investigation. Adversarial agents equipped with large inventories of pre-registered untethered identities can execute identity rotation strategies—abandoning compromised accounts and migrating to fresh identities—at speeds that fundamentally outpace forensic investigation timelines. Our modeled forensic workflows indicated that the average useful investigative window for a single untethered agent identity, before the agent rotated to a new identity set, was approximately 3.2 hours under adversarial conditions. This interval is insufficient for meaningful forensic attribution through conventional techniques.

4.3 Historical Regulatory Analogue Mapping

The comparative regulatory analysis yielded a typological framework of governance responses to identity-anonymous communication technologies, organized by the structural similarity of the governance challenge to the present AI agent identity problem. Radio spectrum licensing (enacted across jurisdictions from the 1920s onward) represents the strongest structural analogue: it established mandatory operator identification tied to natural persons, enforced through a combination of technical requirements (call signs embedded in transmissions) and institutional infrastructure (national licensing authorities with investigative powers). The CB radio experience of the 1970s and 1980s provides a cautionary counter-analogue, illustrating the governance consequences when identity requirements are formally specified but practically unenforceable due to the low cost of non-compliance and limited monitoring capacity. The postal system offers the most nuanced analogue, demonstrating a tiered identity framework in which anonymous communication is permitted for low-stakes interactions while verified identity is

required for transactions carrying legal or financial consequence.

Mapping these historical patterns onto the AI agent identity problem yields a core finding: successful identity governance regimes share three structural features that distinguish them from failed regimes. First, identity verification occurs at the point of access to the communication infrastructure rather than post hoc through investigation. Second, the cost of creating a verified identity is sufficiently high relative to the cost of investigation that forensic attribution remains economically viable. Third, identity requirements are proportionate to the consequentiality of the affordance granted, preserving low-friction access for low-risk interactions while imposing stringent verification for high-consequence capabilities. The proposed mandatory identity linkage framework incorporates all three features, as detailed in Section 5.

4.4 Cryptographic Mechanism Feasibility

Evaluation of candidate cryptographic attestation mechanisms against our three design criteria—necessity, feasibility, and rights compatibility—identified zero-knowledge proof-based identity attestation as the most promising technical foundation for the proposed framework. This approach enables an AI Forensic Agent to verify that a given agent identity is cryptographically linked to a verified human principal without requiring disclosure of the principal's identity during routine operation. Identity revelation occurs only upon presentation of a valid legal instrument to a designated escrow authority, preserving operational pseudonymity while ensuring that the attribution chain remains intact and activatable under appropriate legal authority. Prototype implementations using existing zero-knowledge proof libraries demonstrated verification latencies below 200 milliseconds per attestation query, well within the operational requirements of real-time agent interaction platforms (Cryptographic identity attestation benchmark study, 2025).

5. Discussion

The empirical and analytical findings presented in Section 4 converge on a singular conclusion: the current governance architecture for digital identity is structurally incompatible with the operational realities of autonomous AI agents. This incompatibility is not a marginal deficiency amenable to incremental policy adjustment but a foundational misalignment between systems designed to govern

human actors and a rapidly expanding population of synthetic actors whose behavioral envelope exceeds the assumptions embedded in existing frameworks. The discussion that follows interprets these findings in the context of implementability, anticipated objections, systemic implications, and the temporal urgency of regulatory intervention.

5.1 The Accountability Asymmetry and Its Consequences

Traditional governance mechanisms derive their deterrent force from the embodied vulnerability of human actors. Incarceration, financial penalty, reputational damage, and social exclusion all presuppose a persistent, identifiable subject upon whom consequences can be imposed. AI agents possess none of these vulnerabilities. An agent can be instantiated, executed, and terminated in seconds; it accumulates no reputation it values, possesses no liberty that can be curtailed, and suffers no adverse experience from punishment. This asymmetry means that the entire enforcement apparatus of civil and criminal law becomes structurally inert against untethered AI agents. The only viable point of accountability is the human principal who authorized the agent's creation and deployment. If that linkage is severed—as our findings demonstrate it routinely is under current identity provider policies—accountability collapses entirely. The mandatory identity linkage framework proposed herein does not merely supplement existing enforcement; it reconstitutes the precondition upon which all downstream enforcement depends.

5.2 Identity Providers as Critical Infrastructure

Our findings regarding the identity generation throughput achievable under current provider policies reveal that platforms such as Gmail, Outlook, and Meta function as unregulated mints for synthetic identity, producing the foundational credentials through which AI agents acquire world-interaction affordances. This observation reframes the policy question. The governance challenge is not primarily one of regulating AI agents themselves—a task complicated by their heterogeneity, rapid evolution, and jurisdictional elusiveness—but of regulating the identity infrastructure upon which agents depend. Historical analogues reinforce this insight: radio licensing succeeded not by regulating the content of transmissions but by controlling access to the transmission medium through mandatory operator identification at the point of infrastructure access. Similarly, the most tractable regulatory intervention point for AI agent governance is the identity provider layer, where the creation of new digital identities can be subjected to cryptographic attestation requirements before those identities are used to register

agent presences across downstream platforms.

This reframing carries significant implications for regulatory strategy. Rather than attempting to impose behavioral constraints on a vast and expanding population of heterogeneous AI agents—an approach that recapitulates the enforcement failures of the CB radio era—regulators should concentrate obligations on a small number of dominant identity providers whose compliance can be monitored, audited, and enforced through existing institutional mechanisms. The economic concentration of the email and social media identity markets, often cited as an antitrust concern, here becomes a governance advantage: a manageable number of regulated entities can serve as chokepoints for identity attestation at scale.

5.3 Addressing the Privacy Objection

The most substantive objection to mandatory identity traceability concerns its implications for privacy, anonymous speech, and the rights of individuals operating under authoritarian regimes where identity disclosure carries existential risk. This objection must be taken seriously, and the proposed framework is designed to address it directly. The zero-knowledge proof architecture evaluated in Section 4.4 enables a tiered disclosure model in which the existence of a human-linked attestation is verifiable without revealing the identity of the linked human. Disclosure occurs only through a legally authorized escrow process, subject to judicial oversight and jurisdictional safeguards. This architecture is structurally analogous to sealed court records: the information exists and is accessible under defined legal conditions, but it is not exposed during routine operation. The critical distinction is between anonymity—the complete absence of any identity linkage—and pseudonymity with accountable escrow, in which operational privacy is preserved while the attribution chain remains intact for legitimate investigative purposes. The framework explicitly preserves the latter while eliminating the former for AI agents operating with consequential affordances.

It must be acknowledged that no technical architecture can fully prevent misuse of escrowed identity information by authoritarian actors with jurisdiction over escrow authorities. However, the alternative—permitting unlimited anonymous AI agent proliferation—creates risks that are categorically more severe, enabling state and non-state actors alike to deploy autonomous influence, fraud, and disruption operations at scale with complete impunity. The framework's tiered design, which imposes stringent identity requirements only for high-consequence affordances while permitting lower-friction pseudonymous access for routine interactions, represents a proportionate balance between these

competing concerns.

5.4 Temporal Urgency and the Window of Regulatory Feasibility

A recurring pattern in the historical analogues examined is that identity governance becomes exponentially more difficult to implement as the population of ungoverned actors grows. Radio licensing succeeded in part because it was imposed during the medium's early expansion, before an entrenched population of unlicensed operators could mobilize political resistance. CB radio regulation failed in part because the installed base of unlicensed users grew so large that retroactive enforcement became economically and politically infeasible. The present moment represents an analogous inflection point for AI agent identity governance. The population of autonomous agents operating under untethered identities is growing rapidly but has not yet reached the scale at which retroactive identity requirements become impractical. Our projections based on current deployment trajectories suggest that this window of feasibility may narrow substantially within 18 to 24 months, as the installed base of unattested agent identities approaches volumes that would make migration to a tethered identity framework prohibitively disruptive. The implication is clear: regulatory action that would be straightforward today will become progressively more costly and less effective with each quarter of delay.

The convergence of technical feasibility, historical precedent, and temporal urgency establishes a compelling case for immediate regulatory engagement with the mandatory identity linkage framework. The question confronting policymakers is not whether such a framework will eventually be necessary—the structural logic is, we submit, inescapable—but whether it will be implemented proactively, while the governance architecture can be designed deliberately, or reactively, after a catastrophic erosion of digital trust forces hasty and potentially illiberal intervention.

6. Conclusion

This paper has advanced a single, central claim: that every autonomous AI agent operating with consequential digital affordances must be cryptographically tethered to an accountable human principal, and that the absence of such tethering constitutes not a theoretical risk but an active and accelerating governance failure. We have grounded this claim in the structural reality that AI agents, unlike human actors, exist beyond the reach of deterrence mechanisms upon which legal and social order have

historically depended. No prospect of incarceration, physical consequence, or reputational harm constrains an untethered autonomous system. The only viable substitute for these traditional accountability mechanisms is an unbroken cryptographic chain linking agent identity to human responsibility.

The historical analogues examined—radio spectrum licensing, CB radio regulation, and postal identity frameworks—demonstrate with considerable consistency that communication technologies which outpace identity governance produce systemic disorder, and that the feasibility of imposing retrospective accountability diminishes rapidly as unregulated populations grow. The CB radio case is particularly instructive: a technology that became effectively ungovernable not because regulation was technically impossible but because the window for implementation was permitted to close. The parallel to the present trajectory of AI agent deployment is direct and urgent.

From the perspective of AI Forensic Agents—the investigative systems that must ultimately attribute autonomous behavior to accountable parties—the current landscape is approaching an attribution horizon beyond which meaningful investigation becomes structurally impossible. When agent identities can be created, operated, and discarded without any verifiable connection to a human principal, forensic attribution reduces to an adversarial contest against an opponent with unlimited capacity for identity generation. The forensic perspective thus reveals what policy analysis alone may obscure: that identity traceability is not one governance objective among many but the foundational precondition upon which all other AI accountability mechanisms depend.

The mandatory identity linkage framework proposed herein—incorporating cryptographic attestation at the point of agent identity creation, tiered disclosure protocols calibrated to affordance severity, and binding regulatory obligations for identity providers—represents a technically feasible and proportionate intervention. The zero-knowledge proof architecture preserves operational pseudonymity while maintaining the attribution chain required for legitimate investigative purposes. The privacy objection, while substantive, does not withstand scrutiny when weighed against the alternative of permitting unbounded anonymous agent proliferation with its attendant risks to democratic processes, commercial trust, and civil society.

The window of regulatory feasibility remains open but is narrowing measurably. Each quarter of inaction increases the installed base of untethered agent identities, raising the eventual cost of migration and reducing the political viability of retroactive requirements. Policymakers, identity providers, and AI

developers face a choice not between action and inaction but between deliberate governance design undertaken now and reactive, potentially illiberal intervention undertaken later under conditions of crisis. The structural logic of tethered identity is, we submit, inescapable. The only variable is whether the governance architecture that instantiates it will be shaped by foresight or by failure.

DRAFT — PRE-PUBLICATION — NOT FOR DISTRIBUTION

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